

Quiz 3: Conditional Expectation, Entropy, Stochastic Processes

Homework for lectures 5 and 6. Notation as in the lectures. Twelve problems, numbered 1–12 across the three parts, four per part; each needs the concept of its part.

Part A: Conditional Expectation

Notation as in the Conditional Expectation lecture: two dice Y (first) and Z (second), $X = Y + Z$ and $D = Y - Z$; the rows of the two-dice table are the events $\{Y = y\}$.

1. A supplier's defect rate is unknown: model it as Θ uniform on $[0, \frac{1}{5}]$. Given $\Theta = \theta$, the number D of defective items in a sample of 10 is binomial: $\mathbb{E}[D \mid \Theta = \theta] = 10\theta$ and $\mathbb{E}[D^2 \mid \Theta = \theta] = 10\theta(1 - \theta) + 100\theta^2$. Compute $\mathbb{E}[D]$ and $\text{Var}(D)$, and compare $\text{Var}(D)$ with the variance $10 \cdot 0.1 \cdot 0.9$ of a binomial with the average defect rate 0.1. Then compute $\mathbb{P}(D = 0)$ directly, by integrating over θ , and compare with 0.9^{10} .
2. Toss a fair coin until the pattern THH first appears; then the same for THT and for TTT. Compute the three expected waiting times by conditioning on the first toss from each of the three states “no progress”, “T”, “TH” (respectively “TT”). Which pattern arrives soonest, and why are three patterns of the same length not equally quick?
3. Two dice, $D = Y - Z$, $X = Y + Z$. (a) Show that $\mathbb{E}[D \mid X] = 0 = \mathbb{E}[D]$ as random variables, by considering how the diagonal $\{X = x\}$ lies relative to the main diagonal. (b) Show that $\mathbb{E}[X \mid D] = 7 = \mathbb{E}[X]$ (which cells form $\{D = d\}$, and what is X on them?). Deduce $\text{Cov}(X, D) = 0$. (c) Show that X and D are nevertheless *not* independent, by comparing the parities of X and D or by computing $\mathbb{P}(D = 5, X = 7)$ against $\mathbb{P}(D = 5)\mathbb{P}(X = 7)$. Which result of the lecture does this illustrate?
4. Let $X_1, X_2, \dots$ be independent, each uniform on $\{1, \dots, 6\}$, $\mu = \mathbb{E}[X_i] = 3.5$, and let N be the index of the first roll showing 5 or 6; N is determined by the X_i and not independent of them. (a) Let $S = X_1 + \dots + X_N$ include the stopping roll. Compute $\mathbb{E}[S]$ by conditioning on the first roll, and compare with $\mu \mathbb{E}[N]$. (b) Let $S' = X_1 + \dots + X_{N-1}$ exclude it. Compute $\mathbb{E}[S']$ the same way and compare with $\mu \mathbb{E}[N - 1]$. Explain the two comparisons.

Part B: Entropy

Logarithms are base 2, the unit is the bit, and $0 \log 0 = 0$. Notation as in the Entropy lecture: p_X is the pmf of X ; $H(X)$ is entropy and $h(p)$ the binary entropy function; $L(c)$ is the average length of the prefix-free code c and L^* the optimal average length; $H(P, Q)$ is the cross-entropy of the truth P with the code or model Q , $D(P \parallel Q) = H(P, Q) - H(P)$; $H(X \mid Y)$ and $I(X; Y)$ are conditional entropy and mutual information.

5. Optimality of the rover's code. Let $p = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$. (a) Using Kraft's inequality and Gibbs' lemma, show that every prefix-free code for four symbols has $L(c) \geq 1.75$ under p , so 0, 10, 110, 111 is optimal. (b) Show that the only length profiles for four symbols with Kraft sum exactly 1 are (2, 2, 2, 2) and (1, 2, 3, 3), up to order. (Consider the shortest

- length.) (c) Take as given that an optimal code has Kraft sum 1 (otherwise a codeword can be shortened). Deduce that for *every* pmf on four symbols the optimal code is either the fixed-length code or 0, 10, 110, 111 with the symbols assigned in decreasing order of probability, whichever has the smaller L . (d) Apply (c) to $p = (0.45, 0.25, 0.15, 0.15)$ and to $p = (0.3, 0.3, 0.2, 0.2)$, and compare L^* with H in each case. Why can the source coding theorem alone not decide which code is optimal here?
6. Coin A shows heads with probability 0.9, coin B is fair. One coin is chosen uniformly at random ($C \in \{A, B\}$) and tossed once, giving $X \in \{H, T\}$. (a) Compute the pmf of X and $H(X)$. (b) Compute $H(X | C = A)$, $H(X | C = B)$ and $H(X | C)$. (c) Compare $H(X)$ with $H(X | C)$ and state the property of h that this illustrates. What is $I(X; C)$? (d) Compute $\mathbb{P}(C = A | X = H)$ and $\mathbb{P}(C = A | X = T)$, then $H(C | X)$, and check $H(C) - H(C | X) = I(X; C)$.
7. The rover has two missions, equally likely: $P_1 = (\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$ and $P_2 = (\frac{1}{8}, \frac{1}{8}, \frac{1}{4}, \frac{1}{2})$ for (up, down, left, right). One code, built for a pmf Q , must be installed before the mission is known; its cost is $\frac{1}{2}H(P_1, Q) + \frac{1}{2}H(P_2, Q)$ bits per instruction. (a) Show that this cost equals $H(\bar{P}, Q)$ with $\bar{P} = \frac{1}{2}(P_1 + P_2)$, and compute $\bar{P}$. (b) Find the Q that minimises the cost and the minimal cost. Compare with keeping the old code ($Q = P_1$), with the fixed-length code, and with the 1.75 bits achievable if the mission were known. (c) Show that the average waste $\frac{1}{2}D(P_1 \| \bar{P}) + \frac{1}{2}D(P_2 \| \bar{P})$ equals $I(M; X)$, the mutual information between the mission $M \in \{1, 2\}$ and one instruction X , and compute it. (d) Codeword lengths must be integers. Using Problem 5(b), find the best single-symbol code for the two-mission problem. What must be done to approach the cost in (b)?
8. A binary text source is modelled as a Markov chain on $S = \{0, 1\}$ with transition matrix $Q = \begin{pmatrix} 0.9 & 0.1 \\ 0.3 & 0.7 \end{pmatrix}$. (a) Find the stationary pmf π . (b) Compute the entropy rate $H(X_{i+1} | X_i)$ and the monographic entropy $H(\pi)$, and identify their difference. (c) Compute $H(X_1, X_2, X_3)$ for the chain started from π , once by the chain rule and once by listing the eight probabilities. (d) Give the perplexity of the Markov model, of the monographic model and of the uniform model. (e) What does a compressor built on the monographic model pay per symbol on this text, and how much of it is waste?

Part C: Stochastic Processes

Notation as in the Stochastic Processes lecture: B is a standard Brownian motion; $\mathbb{P}_i, \mathbb{E}_i$ refer to a start at i ; $p = 1 - q$ is the probability of winning a round, $r = q/p$. $\mathbf{P} = (p(i, j))$ is the transition matrix, row i the conditional pmf of the next state given the current state i ; π is a stationary distribution; $h_i = \mathbb{P}_i(\text{reach the target})$ and $k_i = \mathbb{E}_i[\text{time to absorption}]$ are hitting probabilities and times. Φ is the $N(0, 1)$ cdf; values used below: $\Phi(0.100) = 0.540$, $\Phi(0.500) = 0.691$, $\Phi(0.674) = 0.750$, $\Phi(1.000) = 0.841$.

9. The reflection principle for Brownian motion: $\mathbb{P}(\max_{s \leq t} B_s \geq a) = 2\mathbb{P}(B_t \geq a)$. Let T_1 be the first time B reaches 1. (a) Compute $\mathbb{P}(\max_{s \leq 4} B_s \geq 2)$. (b) Write $\mathbb{P}(T_1 \leq t)$ in terms of Φ and evaluate it at $t = 1, 4, 100$. (c) Find the median of T_1 (the t with $\mathbb{P}(T_1 \leq t) = \frac{1}{2}$). (d) By scaling, what is the median of T_a ? (e) The lecture showed $\mathbb{E}[T_1] = \infty$. Reconcile this with a median of about 2.
10. Bold play. $p = 0.4$, fortune $\mathcal{L}3$, target $\mathcal{L}10$. (a) Timid play ($\mathcal{L}1$ per round): compute h_3 and $\mathbb{E}_3[T]$ from the lecture's ruin formulas. (b) Bold play: each round stake $\min(i, 10 - i)$, so the fortune moves from 3 to 6 or 0, from 6 to 10 or 2, from 2 to 4 or 0, from 4 to 8 or 0, from 8 to 10 or 6. Draw the chain on $\{0, 2, 3, 4, 6, 8, 10\}$ and write the hitting system

- for $h_i = \mathbb{P}_i(\text{reach } 10)$. (c) Solve it and compare h_3 with (a). (d) Compute the expected number of bets under bold play from $\mathcal{L}3$. (e) Repeat (c) for the fair game $p = \frac{1}{2}$. Why is the answer the same as for timid play there?
11. PageRank with damping $d = 0.85$. (a) Three pages: $A \rightarrow B$, $A \rightarrow C$, $B \rightarrow C$, $C \rightarrow A$. Write $\mathbf{H}$ and $\mathbf{P} = 0.85\mathbf{H} + 0.05\mathbf{J}$. (b) Solve $\pi\mathbf{P} = \pi$ and rank the pages. A and B each have exactly one in-link; why does A rank higher? (c) Find π without damping ($d = 1$). (d) Add a page D with links $C \rightarrow D$ (so C now links to A and D) and $D \rightarrow D$ only. Find π without damping, and with $d = 0.85$. What does D do to the surfer, and how much does damping help?
12. Tennis at deuce. From deuce the server wins each point with probability p , independently. States: D (deuce), I (advantage server), O (advantage receiver), W (game won), L (game lost). From D the next state is I or O; from I it is W or D; from O it is D or L. (a) Write the transition matrix. (b) Write the hitting system for $h_i = \mathbb{P}_i(\text{reach } W)$ and solve it; show $h_D = \frac{p^2}{p^2 + q^2}$. (c) Write the system for $k_i = \mathbb{E}_i[\text{points until the game ends}]$ and show $k_D = \frac{2}{1 - 2pq}$. (d) Evaluate h_D and k_D for $p = 0.5, 0.55, 0.6, 0.7$. What does a 55/45 edge per point become per game? (e) Express h_D as an infinite series by summing over the number of points played, and explain why the first-step system avoids the series.